Jacob Zheng

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Education

University of Waterloo *Waterloo, ON*
Bachelor of Mathematics, Honours, Co-op

Experience

Software Developer (Drupal), Incoming Co-op Jan. 2027 – Apr. 2027
Agriculture and Agri-Food Canada (FSWEP) *Ottawa, ON*

Software Engineer Sep. 2025 – Aug. 2026
Waterloo Reality Labs *Waterloo, ON*

- Built a **gesture recognition pipeline** on the **Meta XR All-in-One SDK**, classifying **5+ custom gestures** unsupported by the base SDK at **92% accuracy**, enabling new hands-free interactions for lab XR projects.
- Trained a **neural network** in **PyTorch** on **2,000+ labeled** finger curl and flexion samples, tuning input features and thresholds to stay robust across different hand sizes, poses, lighting conditions, and individual users.
- Deployed it to UWRL's standalone XR headset with a **per-user calibration tool**, emitting real-time gesture events at **under 20ms latency** to downstream XR applications and demos built by the rest of the team.

AI Research Assistant Apr. 2026 – Aug. 2026
MDPharma

- Designed a **Just-in-Time Adaptive Intervention (JITAI)** system for a Gen Z wellness app, fusing **3 signal streams** (calendar, biometric, GPS) to infer user state and deliver tailored wellness prompts in real time.
- Benchmarked **10+ candidate models** for user state inference and prompt timing, then selected an approach that balances accuracy, on-device compute cost, and user privacy for a mobile-first consumer product.
- Delivered a working **prototype** and a **25-page implementation guide** spanning model selection, training, and deployment, giving the engineering team a clear, step-by-step roadmap from research to a production launch.

Projects

Toronto Traffic Disruption Simulator Oct. 2026

- Built a **Next.js** / **TypeScript** simulator of road closures across **42,987 Toronto road segments**.
- Rendered traffic in **2D/3D** with **Mapbox GL** and **deck.gl**, modeling congestion via **BPR functions**.
- Built a **one-way-aware rerouting API** using **Gemini** to redistribute diverted traffic.

Multimodal Hands-Free IDE Controller Jul. 2026

- Built the **Python backend** for a **VS Code extension** for coding via **gaze, gestures, and voice**.
- Calibrated gaze with **SVD ridge regression** on **MediaPipe** iris data from **under 50 samples**.
- Designed a **hold-gated gesture state machine** exposed via a **token-authenticated API**.

Photo Authenticity Verification App Jul. 2026

- Built a **React Native** camera app that cryptographically proves photo authenticity at capture.
- Signed photos on-device with **Ed25519** and anchored hashes to **Solana**, with metadata in **MongoDB Atlas**.
- Built a **3-tier verifier** with **SHA-256** and **perceptual hashing**, flagging **6+ byte** hash diffs.

Multimodal AI Teaching Assistant Mar. 2026

- Built a real-time **AI teaching assistant** that detects student disengagement via webcam.
- Detected disengagement with **90% accuracy**, streaming events over **WebSockets** at **sub-75ms latency**.
- Delivered voice tutoring with **ElevenLabs** and **Gemini** on **FastAPI** and **Next.js**.

Skills

Languages: Python, TypeScript, JavaScript, SQL, C, C++, Bash
Frameworks & Libraries: React, Next.js, Node.js, FastAPI, PyTorch, PostgreSQL, MongoDB, MediaPipe
Cloud & Tools: Microsoft Azure (AZ-900, AI-900), AWS (AI Practitioner), Docker, Git, Linux